



DECK DEPARTMENT TRAINING

Cargo Securing

Principles, equipment and procedures for securing cargo safely on general cargo and multipurpose vessels

SOLAS Ch. VI & VII

CSS Code

CSM

Company SMS



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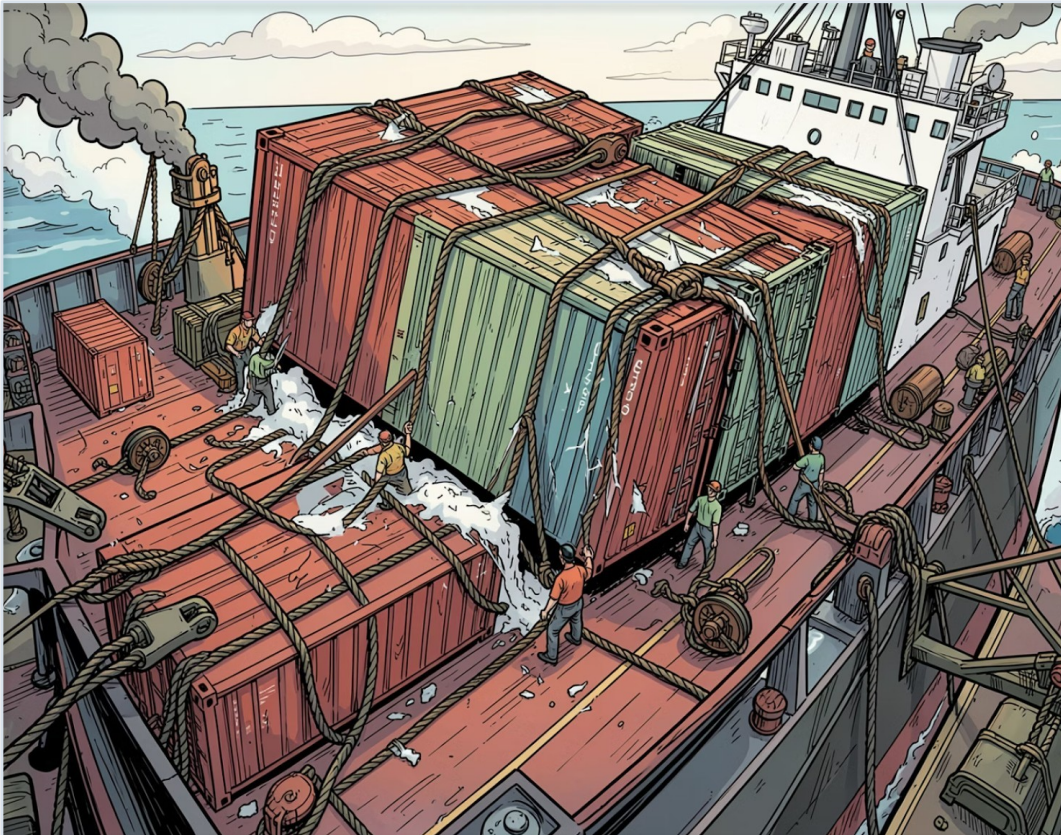
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Why Cargo Securing Matters

Improper stowage or insufficient securing is one of the leading causes of serious incidents at sea. The consequences can escalate rapidly.



Cargo Shifting

Loss of stability and structural damage from cargo moving within the hold or on deck.



Crew Injury or Fatality

Parted lashings and shifting loads are a leading cause of serious injury on deck.



Pollution & Vessel Loss

In extreme cases, cargo incidents lead to hull breach, pollution or total loss.



Financial & Reputational Cost

Example: a single shifted steel coil can cause hull damage, cargo claims and off-hire.

Forces Acting on Cargo at Sea

A vessel's motion at sea generates forces from every direction that a securing arrangement must be able to resist.



Rolling

Side-to-side motion creates the largest lateral forces on cargo and lashings, especially in beam seas.



Pitching

Bow-to-stern motion adds longitudinal force, significant when heading into or running with the sea.



Heaving (Vertical Acceleration)

Vertical acceleration at the ship's extremities can briefly exceed 1g in heavy weather.



Wind & Green Water

Wind pressure and boarding seas add lateral force and can wash away unsecured items.



Forces are greatest at the ship's extremities — forward, aft and upper-deck stows need the most conservative securing arrangements.

Training Objectives

What crew members will be able to do on completion of this training.



Identify Hazards

Recognize risks associated with unsecured or poorly secured cargo.



Use Equipment Correctly

Select and operate chain lashings, turnbuckles, shackles and wire safely.



Follow the CSM

Apply vessel-specific procedures from the Cargo Securing Manual.



Heavy Weather Readiness

Apply precautions before, during and after heavy-weather passages.



Communicate Effectively

Report cargo condition and any shift immediately to the bridge.



Complete Training Records

Ensure drills and sign-offs are logged in the training record.

Roles & Responsibilities

Cargo securing depends on clear responsibility, from the Master's authority down to the crew member tensioning a lashing.

Role	Responsibility
Master	Overall responsibility for safe carriage; authorizes deviation for weather; approves re-securing after cargo movement
Chief Officer	Plans stowage and securing per the CSM; supervises lashing operations; signs training records
Officer of the Watch	Monitors cargo condition during the watch; reports any change immediately to the Master or Chief Officer
Deck Crew	Carries out lashing and unlashings correctly; uses PPE; reports defective equipment
DPA / Shore Management	Reviews non-conformities; ensures the CSM and equipment certificates are kept current

Standards & References

All cargo securing operations must comply with international regulations and company-specific requirements.



Standard	Reference	Applies To
SOLAS Ch. VI & VII	Intl. Convention for the Safety of Life at Sea	Carriage of cargoes & dangerous goods — the international baseline
CSS Code	IMO Code of Safe Practice for Cargo Stowage and Securing	Securing arrangements, lashing calculations and equipment standards
Cargo Securing Manual (CSM)	Vessel-specific manual approved by the flag state	Approved lashing patterns, cargo groupings and equipment SWL
Company SMS	Safety Management System procedures	Risk assessment, permit to work and non-conformity reporting
Flag State / Class	National regulations and classification society rules	Manual approval and equipment certification requirements

The Cargo Securing Manual

The CSM is the vessel's primary reference for how cargo is stowed and secured — every deck officer must know its structure.



General Information — Vessel particulars, the cargo securing devices register, and approval details.



Standard Cargo Securing Conditions — Pre-approved arrangements for typical cargo types and stow positions.



Non-Standardized Cargo — Cargo not covered by a standard condition requires a case-by-case calculation.



Securing Devices & SWL Register — An inventory of all fittings on board and their safe working loads.



Maintenance & Renewal — Periodic inspection and replacement criteria for lashing equipment.

Cargo Types & Securing Methods

Each cargo type carried on a general cargo vessel requires a tailored securing arrangement. One method does not fit all.



Steel Coils & Plates

Require cradles, timber dunnage and chain lashings to prevent rolling.



Timber & Pipes

Long, cylindrical or bundled cargo is prone to shifting — stanchions and wires are essential.



Heavy Lift & Project Cargo

Custom lashing plans are required; load distribution on deck must be verified.



Palletized Cargo

Stretch-wrap, strapping and lashing bars prevent toppling in holds and on deck.



Containers (Non-Cellular)

Twist-locks, lashing rods and bridge fittings must match the approved plan.



Drums, Reels & IBC Containers

Chocking, banding and drip trays prevent movement and contain leaks.

Container Securing on Deck

Where containers are carried on general cargo vessels, securing follows the vessel's approved lashing plan exactly.



Twist Locks — Semi-automatic or manual fittings connecting container corner castings to the deck or to the container below.



Lashing Rods & Turnbuckles — Diagonal rods from upper corners to deck sockets or bridge fittings, tensioned evenly on both sides.



Corner Castings & Bridge Fittings — The load path between containers and ship structure — must be free of cracks or deformation.



Stack Weight & Height Limits — Never exceed the approved stack weight or lashing pattern for the stow position.



Securing Sequence — Lash from the bottom tier upward; check tension again after the first few hours at sea.

Timber Deck Cargo — Special Requirements

Timber carried as deck cargo is covered by its own IMO code, in addition to the CSS Code.



Code Application — The IMO Code of Safe Practice for Ships Carrying Timber Deck Cargoes applies when timber is carried on deck.



Uprights & Stanchions — Fitted at sufficient spacing and strong enough to resist the load of the stow.



Lashing Checks — Tensioned and re-checked regularly, particularly within the first 24 hours at sea.



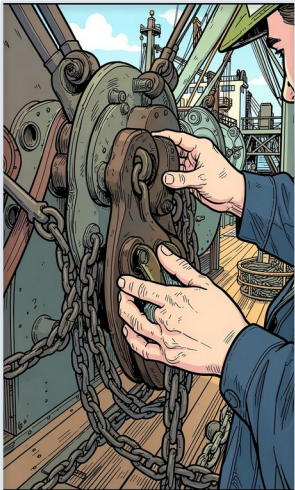
Access & Visibility — Deck cargo must never obstruct navigation, escape routes or venting arrangements.



Timber Load Line — Applies only when the cargo meets the Code's stowage and securing conditions.

Cargo Securing Equipment

Know the equipment, check it before every use, and reject anything that does not pass inspection.



Equipment Types

- Chain & wire lashings
- Turnbuckles & shackles
- D-rings & lashing points
- Dunnage & timber wedges



Before Every Use

- Check SWL markings
- Inspect for cracks or corrosion
- Report defective equipment
- Store properly after use



Common Defects — Reject

- Cracked or bent shackle
- Corroded, stretched chain link
- Frayed or kinked wire
- Cut or UV-damaged strap



Never use securing equipment without confirming that its SWL is appropriate for the cargo load being applied.

Securing Devices & Certification

Every securing device on board falls into one of two categories, and both must be marked, inspected and certified.



Fixed Fittings — Pad eyes, D-rings and lashing points welded to deck or hatch covers; part of ship structure, checked at class surveys.



Portable Devices — Chain lashings, wire lashings, turnbuckles and twist locks carried as loose equipment, each marked with its SWL.



Marking & Certification — Every device should show its SWL and, where applicable, a certificate or maker's data plate.



Periodic Load Testing — Devices are subject to periodic proof-load testing per maker or class requirements, logged in the equipment register.



Withdrawal Criteria — Any device that is cracked, deformed, corroded beyond limit, or missing its marking must be withdrawn immediately.

Calculating Securing Requirements

The basic principle behind every approved arrangement in the CSM, in plain terms.



Worst Expected Conditions — The arrangement must resist the forces of the worst sea state expected for the voyage and stow position.



Safety Factor Applied — A safety factor always separates the calculated force from a device's Maximum Securing Load (MSL).



MSL Convention — MSL is typically taken as 50% of a new device's breaking load, or as stated in the CSM or by the maker.



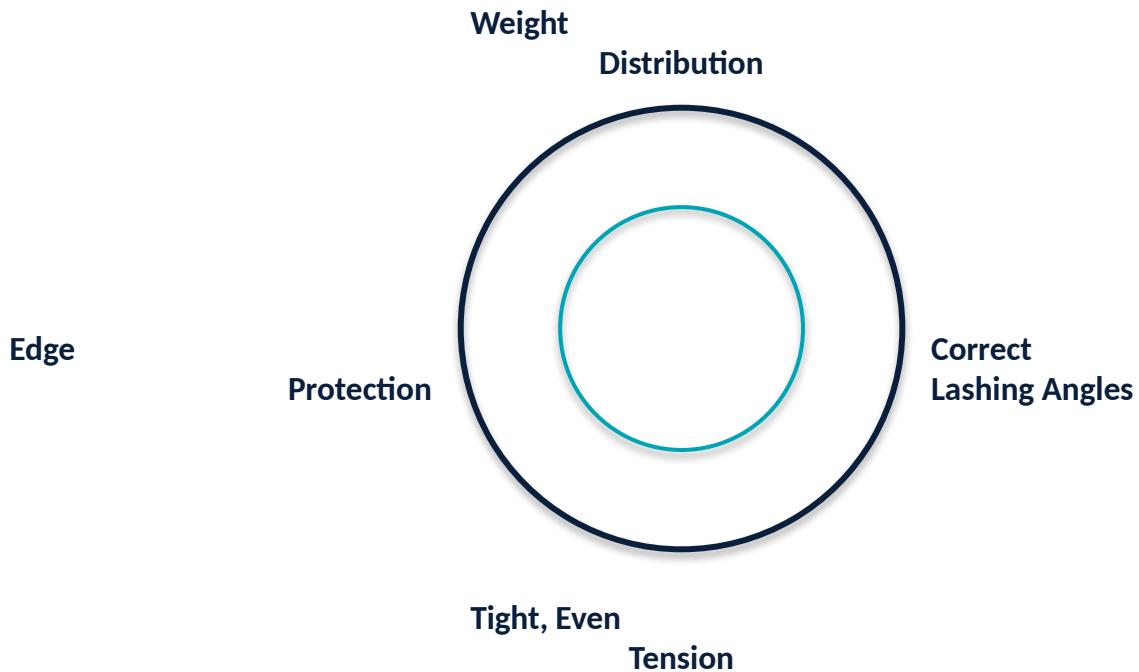
Effective Lashing Angles — lashings share vertical and horizontal restraint — very shallow or very steep angles reduce effectiveness.



Follow the CSM: when in doubt, use the vessel-specific arrangement in the manual rather than an improvised method. An arrangement approved for a stow position must never be reduced without a new calculation.

Principles of Safe Cargo Securing

Applying all four principles together ensures cargo cannot shift during rolling, pitching or combined vessel motions at sea.



All four principles apply together for every lashing arrangement.

Personal Safety During Operations



PPE Required

Hard hat, gloves and safety boots at all times.



Snap-Back Zones

Stay clear — a parted lashing can be fatal.



Fall Protection

Use a harness when working at height on deck cargo.



Permit to Work

Follow risk assessments and PTW for all lashing work.

Heavy Weather & Voyage Inspections

Cargo securing must be checked on a fixed schedule, and tightened ahead of deteriorating weather.

Heavy Weather Checklist

1

Before
Tighten all lashings; add additional securing where needed; secure loose deck equipment.

2

After
Inspect cargo condition; verify lashing tension; report any cargo movement to the bridge.

Voyage Inspection Schedule

When	Inspection Required
Before departure	Full lashing check
After departure	Confirm no movement in port
Daily at sea	Visual round of all securing
Before heavy weather	Tighten and add lashings
After heavy weather	Full condition assessment
Prior to port arrival	Confirm no shift before cargo ops

Weather Routing & Voyage Planning

The best cargo securing outcome is avoiding the worst of the weather in the first place.



Voyage Planning — Accounts for seasonal weather patterns and avoids known areas of heavy weather where practicable.



Daily Weather Review — Forecasts and routing advice are reviewed daily and shared with the deck team.



Master's Authority — The Master has full authority to alter course, speed or heading to reduce forces on cargo, regardless of schedule.



Proactive Securing — Additional lashings are applied before entering a forecast area of deteriorating weather, not after.



Log Every Deviation — Any deviation made for weather is logged in the deck log with the reasons recorded.

Emergency Response: Cargo Movement Detected

Unusual noise, an unexplained list, or visible shift on deck or CCTV all indicate cargo movement — act on the sequence below.

1

Inform the Bridge

Alert the Officer of the Watch immediately — do not delay.

2

Restrict Access

Secure the area; prevent crew from entering the unsafe zone.

3

Assess Condition

Evaluate the extent of the cargo shift from a safe position.

4

Reduce Vessel Motion

The Master may alter course or speed to minimize forces.

5

Correct Under Safe Conditions

Re-secure only when authorized by the Master and conditions allow.

6

Record & Report

Log time, position, weather and actions; report as an SMS non-conformity.



Crew safety is the absolute first priority. No cargo is worth risking a life.

Reporting, Records & Common Non-Conformities

Good recordkeeping closes the loop — and knowing common findings helps a vessel stay ahead of them.

Recordkeeping

- Training records signed by the Chief Officer after every session
- Equipment register updated at each inspection
- CSM amendments approved by the flag state or class before use
- Non-conformities reported through the Company SMS

Common Findings During Inspections

- Missing or illegible SWL markings
- CSM not readily available on the bridge or on deck
- Lashing equipment corroded or beyond its service life
- Securing arrangement not matching the approved CSM
- Training records incomplete or unsigned

Key Takeaways & Assessment

What Every Crew Member Must Remember

Know Your Equipment

Inspect before use; never exceed SWL; report defects immediately.

Inspect Regularly

Daily rounds and pre- and post-heavy-weather checks are mandatory.

Understand the Forces

Know why rolling, pitching and heaving all add force to an arrangement.

Follow the CSM

The Cargo Securing Manual is your primary reference — use it every time.

Act Immediately

If cargo moves, inform the bridge first and act only when it is safe.

Escalate When Unsure

If cargo is not covered by the CSM, stop and get it calculated — never improvise.

Crew Assessment Questions

1. Why is proper cargo securing critical to vessel safety?
2. What are the main risks of cargo shifting at sea?
3. What must be checked before using chain lashings?
4. Why are snap-back zones dangerous?
5. When must cargo inspections be carried out during a voyage?
6. What steps should be taken before heavy weather?
7. What is the correct response when cargo movement is observed?
8. What is the Master's authority regarding weather deviation for cargo safety?
9. Who must sign the training record after each session?



Training records must be completed and signed by the Chief Officer following each session.